

CRITERION 7	Continuous Improvement	50
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7 CONTINUOUS IMPROVEMENT (50)

7.1. Actions taken based on the results of evaluation of each of the POs & PSOs (20)

Identify the areas of weaknesses in the program based on the analysis of evaluation of POs & PSOs attainment levels. Measures identified and implemented to improve POs & PSOs attainment levels for the assessment years.

Actions to be written as per table in 3.3.2.

POs & PSOs attainment levels and Actions for improvement – CAYm (2023 – 2024)

POs	Target Level	Attainment Level	Observations
PO1: Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.			
PO1	1.6	2.58	➤ Target attained ➤ 91% of courses attained the target level by direct assessment (courses attained the target level+ courses attained less than target level - if the attainment is more than 90% of its mapped level) ➤ Attainment by direct assessment is 1.98 & by indirect assessment is 0.6
Action 1: Bridge course were conducted before the commencement of first year to enhance the prerequisite knowledge in Engineering. Action 2: Special Coaching classes were conducted for the courses C301, C310, C403 in order to make the slow learners understand the topics much better. Action 3: For better understanding of the basics concepts in core subjects, students are being taught lab experiments along with the theory subjects.			
PO2: Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.			
PO2	1.6	2.13	➤ Target attained ➤ 83% of courses attained the target level by direct assessment (courses attained the target level+ courses attained less than target level - if the attainment is more than 90% of its mapped level) ➤ Attainment by direct assessment is 1.53 & by indirect assessment is 0.6

Action 1: Tutorials has been planned and given to the students for solving more problems in the course C310. Action 2: Practical sessions were conducted for the course C403 for better understanding of the concepts. Action 3: More number of problems were asked in assignment and the monitoring of the same on a regular basis for the course C408. Action 4: Additional contents are being taught apart from the syllabus to analyse complex problems.			
PO3: Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.			
PO3	1.6	1.93	➤ Target attained ➤ 81% of courses attained the target level by direct assessment (courses attained the target level+ courses attained less than target level - if the attainment is more than 90% of its mapped level) ➤ Attainment by direct assessment is 1.33 & by indirect assessment is 0.6
Action 1: Students are being given design projects which will enhance their design capacity. Action 2: Special classes were conducted to improve the student skills in complex problems for the problematic courses C201, C312 in order to improve the overall course outcome. Action 3: Faculty has been advised to give application-based assignments to students in the course C408. Action 4: Workshop on “Designing and Detailing of Structures” on 31.08.2023 in order to overall attainment of CO in the course C403.			
PO4: Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.			
PO4	1.6	1.91	➤ Target attained ➤ 78% of courses attained the target level by direct assessment (courses attained the target level+ courses attained less than target level - if the attainment is more than 90% of its mapped level) ➤ Attainment by direct assessment is 1.31 & by indirect assessment is 0.6
Action 1: Conducted cube contest from ICI in association with Ramco Cements on 06.10.2022. Action 2: Content Beyond/Additional Experiments were included in laboratory courses. Action 3: Student projects are being reviewed by external industrial expert to improve the overall CO attainment and the quality of the project. Action 4: Motivated students to do L&T Edutech Online course.			
PO5: Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and			

modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.			
PO5	1.6	1.88	➤ Target attained ➤ 79% of courses attained the target level by direct assessment (courses attained the target level+ courses attained less than target level - if the attainment is more than 90% of its mapped level) ➤ Attainment by direct assessment is 1.28 & by indirect assessment is 0.6
Action 1: Hands on Training on “STAAD.Pro Connect Edition V22” from ICI Student chapter from 27.01.2023 to 28.01.2023 was conducted for improving the attainment of C403. Action 2: One-week AU approved VAC on “Project Planning and Management using Primavera” from ICI Student chapter from 25.01.2023 to 30.01.2023. Action 3: Nan Mudhalvan course on Building Information Modelling, Architectural Visualization using AR/VR and Concrete Building Design were conducted to expose knowledge on different software among the students. Action 4: Faculty members underwent training and workshops on modern tool usage to effectively teach students.			
PO6: The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.			
PO6	1.6	2.07	➤ Target attained ➤ 89% of courses attained the target level by direct assessment (courses attained the target level+ courses attained less than target level - if the attainment is more than 90% of its mapped level) ➤ Attainment by direct assessment is 1.47 & by indirect assessment is 0.6
Action 1: Students are being involved in club activities like NSS, Eco club etc., to give engineering solutions to the society. Action 2: Programs are being organized through the Center for Women's Welfare to raise awareness about health and safety. Action 3: One hour per week are being allotted for improving student's Personality and Character Development. Action 4: Real-world projects like Household Waste Composter has been carried out by the students for environmental up gradation and to develop the societal need. Action 5: Students participated in Blood Donation Camps which was organized by NSS. Action 6: Students were made to understand Integrated Solid Waste Management, Handling Separation and storage at source and Incineration through NPTEL videos.			
PO7: Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.			

PO7	1.6	2.02	➤ Target attained ➤ 85% of courses attained the target level by direct assessment (courses attained the target level+ courses attained less than target level - if the attainment is more than 90% of its mapped level) ➤ Attainment by direct assessment is 1.42 & by indirect assessment is 0.6
Action 1: Students are encouraged to analyse the global and Environmental issues to provide solutions in their final year projects in C410 Action 2: Conducted Innovative Practices on Reflective journal on the topic Desalination treatment units for the students to meet the current and future needs considering economic, environmental and sustainable aspects Action 3: Virtual Session on “Green Railway Stations – Sustainable Transport at the heart of Sustainable development goals” was organized on 19.10.2023. Action 4: Virtual Expert Lecture on “A Career in Sustainable Building Industry” on 09.03.2023.			
PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.			
PO8	1.6	1.98	➤ Target attained ➤ 81% of courses attained the target level by direct assessment (courses attained the target level+ courses attained less than target level - if the attainment is more than 90% of its mapped level) ➤ Attainment by direct assessment is 1.38 & by indirect assessment is 0.6
Action 1: Students attended a seminar on “Basics of Vibration Analysis and Seismic Resistant Structures” able to apply ethical principles in real-world scenarios, particularly in the design and construction of RCC and steel structures. Action 2: Students were encouraged to participate in the Cube Contest to arrive the mix proportion by IS code principle. Action 3: Encouraged the students to pursue “Professional Ethics in Engineering” course as elective. Action 4: Students are being followed the code-of conduct in examination process and all our engineering practices.			
PO9: Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.			
PO9	1.6	1.91	➤ Target attained ➤ 79% of courses attained the target level by direct assessment (courses attained the target level+ courses attained less than target level - if the attainment is more than 90% of its mapped level)

			level) ➤ Attainment by direct assessment is 1.31 & by indirect assessment is 0.6
Action 1: The students are being encouraged to organize and lead various technical events to groom their leadership, team work, coordination, and commitment. Action 2: The group of students were encouraged to participate in various events such as Cube Contest, Smart India Hackathon, EUPHORIA' 24 etc., Action 3: Encouraging students to participate in various Co-curricular activities in other colleges and promotion of various clubs and activities. Action 4: Students undergone Industrial Visit in Ramco Cements Limited, Virudhunagar, Surgical Cotton Mills Limited, Rajapalayam and L&T Construction Project – Cochin Dry Dock Site, Kerala to acquire knowledge in Industrial Practice.			
PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.			
PO10	1.6	2.09	➤ Target attained ➤ 93% of courses attained the target level by direct assessment (courses attained the target level+ courses attained less than target level - if the attainment is more than 90% of its mapped level) ➤ Attainment by direct assessment is 1.49 & by indirect assessment is 0.6
Action 1: Students are being encouraged to do morning presentation (8.50 AM – 9.00 AM) during regular working days. Action2: Students are constantly motivated to participate in Conference/Seminar/Workshops/Symposiums. Action 3: Students were given Seminar on the topic like Structural Drafting Using CAD, Preface to ArcGIS etc., Action 4: Students involved in preparation of Newsletter and Magazine. Action5: Soft skill training are being imparted to students to enhance various aspects of communication.			
PO11: Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.			
PO11	1.6	1.98	➤ Target attained ➤ 79% of courses attained the target level by direct assessment (courses attained the target level+ courses attained less than target level - if the attainment is more than 90% of its mapped level) ➤ Attainment by direct assessment is 1.38 & by indirect assessment

			is 0.6
Action 1: Under the TNSCST Student Project Scheme, students received fund for the project “Textile Reinforced Concrete” and they obtained a utilisation certificate for the procurement of materials, testing, and transportation. Action 2: Projects were guided by the faculty members for the benefit of students which incorporates budget, period of project work and factor of delay in project. Action 3: One week AU Approved VAC on “Project Planning and Management using Primavera” from 25.07.2022 to 30.07.2022 was conducted for sharing the concepts of project management skills.			
PO12: Life - long Learning: Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.			
PO12	1.6	1.76	➤ Target attained ➤ 72% of courses attained the target level by direct assessment (courses attained the target level+ courses attained less than target level - if the attainment is more than 90% of its mapped level) ➤ Attainment by direct assessment is 1.16 & by indirect assessment is 0.6
Action 1: Conducted Webinar “GATE CIVIL 2023 – Strategies and Preparation” on 02.02.2023 in association with Pyramid IAS Academy, Karaikudi. Action 2: The resource person from Ramki IAS Academy, Chennai was invited to create Awareness about “UPSC and TNPSC Civil Service Examination” which was organised on 07.02.2023. Action 3: Awareness Program titled “Sustainable Development in Student’s career – Opportunities to study and Work Abroad” was conducted on 03.05.2023. Action 4: Students are being encouraged to do online courses to engage themselves as independent learners. Action 5: Alumni are being invited to share their views and experiences.			
PSO1: Communicate and present civil engineering projects effectively.			
PSO1	1.6	1.78	➤ Target attained ➤ 81% of courses attained the target level by direct assessment (courses attained the target level+ courses attained less than target level - if the attainment is more than 90% of its mapped level) ➤ Attainment by direct assessment is 1.18 & by indirect assessment is 0.6
Action 1: Conducted regular project review sessions to provide constructive feedback on communication and technical presentation skills. Action 2: Students were encouraged to present their projects in Smart India Hackathon.			
PSO2: Use the techniques, skills, and modern engineering tools necessary for civil engineering practice and project management.			

PSO2	1.6	1.97	➤ Target attained ➤ 83% of courses attained the target level by direct assessment (courses attained the target level+ courses attained less than target level - if the attainment is more than 90% of its mapped level) ➤ Attainment by direct assessment is 1.37 & by indirect assessment is 0.6
Action 1: As Civil Engineers rely on complex and sophisticated software to develop designs, and manage projects, workshops like Total station, DGPS etc., were conducted to enhance the students' technical proficiency to meet industry standards. Action 2: Civil engineering profession needs creative real-world-problem-solving skills by generating multiple solutions for major projects. Opportunities were provided to students to showcase their creative ideas to solve the pressing societal issues through projects. Action 3: Hands on Training titled "STAAD.Pro Connect Edition V22" was conducted on 27.01.2023 & 28.01.2023. Action 4: Industrial visits and Internship are being arranged for students in core industries for getting practical exposure.			
PSO3: Provide sustainable solutions to civil engineering problems.			
PSO3	1.6	1.81	➤ Target attained ➤ 78% of courses attained the target level by direct assessment (courses attained the target level+ courses attained less than target level - if the attainment is more than 90% of its mapped level) ➤ Attainment by direct assessment is 1.21 & by indirect assessment is 0.6
Action 1: Technical seminar was conducted on the topic "Alternative Building Technologies" on 04.11.2022. Action 2: Virtual Knowledge Sharing Session was organized in the topic "Integrating Geo-spatial and ML Techniques in Irrigation & Water Resources Engineering" on 06.04.2023 in order to improve the overall attainment of C312.			
PSO4: Perform as design consultants in construction industry for the design of civil engineering structures			
PSO4	1.6	2.05	➤ Target attained ➤ 89% of courses attained the target level by direct assessment (courses attained the target level+ courses attained less than target level - if the attainment is more than 90% of its mapped level) ➤ Attainment by direct assessment is 1.45 & by indirect assessment is 0.6

Action 1: Site visit to the Steel Structures Construction at Surgical Cotton Mills Limited, Rajapalayam was arranged to improve the overall attainment of course outcome in the subject C311.

Action 2: Organized a Workshop titled “Designing and Detailing of Structures” on 31.08.2023 to improve the overall attainment of course outcome in the subject C403.

Action 3: Conducted a Value-added course on “Open Building Designer” from 25.08.2022 to 26.08.2022.

Action 4: Virtual Webinar on the topic “Reality Modelling” was organized on 18.05.2022.

Action 5: Guest Lecture on “Ergonomics in Construction” was conducted on 25.03.2022.

7.2 Academic Audit and Actions taken thereof during the period of Assessment:

The Institution is certified with ISO 9001:2015 Quality management system with process owners, reviewing and approval authorities for 27 academics and supporting procedures. Internal Audit are conducted in line with ISO 9001:2015 (Academic Procedure, Examination Procedure, Training & Placement Procedure, Research & development Procedure etc., and External Surveillance Audit. 45 Faculty members are Certified Internal auditors in our institution from TUV Rhineland. With the support of these auditors, 2 Internal Audits every semester will be conducted and auditors from TUV Rhineland will do annual surveillance audit. One of the important aspects of these audits is **Academic Process** auditing.

ACADEMIC PROCESS AUDIT ASSESSMENT CRITERIA

Is based on Plan-Do-Check- Act Process

Planning Process

- ❖ Subject Preference
- ❖ Subject Allocation
- ❖ Academic Schedule Preparation in line with Anna University Schedule
- ❖ Timetable preparation
- ❖ Course file Preparation
 - Course Plan – (AC 05a)
 - Course Objectives and Course Outcomes, Learning Outcomes, Course Target, CO-PO Matrices, CO-PSO Matrices
 - Lecture Plan, Bridge Course Plan (optional), Content Beyond Syllabus Plan, Tutorial Plan (if applicable), Assignment Plan, Details of e-learning sources (Videos), Mini project plan (if applicable), Innovative Practice plan and Session plan.
 - Course Assessment – (AC 06a)
 - Assessment Methods, Course Assessment Plan, Evaluation Plan and Rubrics for Assessment.
 - Question Bank, Syllabus and other required materials for course handling

This stage will be verified by department HOD, Vice Principal and Approved by Principal.

Do Process

- ❖ **Course delivery** as per plan is executed using **ICT tools** and **Innovative Teaching Methodologies**.
- ❖ Transparent **Assessment Methods** by through Direct and Indirect Assessment are used to know the leaning levels of the students.
 - Direct Assessments- Internal Assessment Tests, Tutorials, Online Quiz, Assignments etc., (to be selected based on course nature)
 - Indirect Assessments - Program Exit Survey, Course Exit Survey etc., (to be selected based on course nature)

Check Process:

- ❖ The conduct of classes is monitored using Class register
- ❖ **Performance** in Internal Assessment test is monitored periodically and analyzed in Class Committee Meeting and Department Review Meeting.
- ❖ **Quality** of Question paper and evaluation is reviewed by HODs and Senior Faculty members.
- ❖ The regular Activities are reviewed by the HODs with the support of ISO 9001:2015 Quality management System.

Act Process:

- ❖ **Preventive and corrective actions** are Planned and executed.
- ❖ **The reforms** in the system, if required is incorporated.

The Quality Management System is Risk Based which has 4 modules

- ✓ Risk Identification
- ✓ Risk Analysis
- ✓ Risk Evaluation
- ✓ Risk Treatment

through which in case of any risk identification related to academic, it will be analysed and accordingly corrective and preventive actions are taken.

ACADEMIC AUDIT FREQUENCY AND CONDUCT MECHANISM:

Table 7.1 Academic Audit Frequency

Sl.No.	Audit Particulars	Audit frequency	Responsibility
1.	Course file verification (Lecture Plan, Assignment Plan, Tutorial Plan, Practical Plan, e-Lecture's Plan etc.,)	Beginning of every Semester	HOD/Principal
2.	Log book verification	Completion of each unit	HOD/Principal
3.	Class register verification	Once in a week	CA/HOD
4.	Continuous Internal assessment question paper verification	Before each Internal Assessment tests	HOD
5.	Internal assessment tests evaluated answer script samples verification	Completion of each Internal Assessment Tests	HOD
6.	Stock verification	One per Year	Internal auditors/ Lead auditor
7.	ISO Internal audit	Two per Semester	Internal auditors/ Lead auditor

8.	Academic audit	One per Year	External Auditor/HOD
9.	External Surveillance audit	One per Year	Lead auditor - TUV Rhineland
10.	ISO certification audit	One per three Years	Lead auditor - TUV Rhineland
11.	IQAC Meeting	Two per Semester	HOD's/ Principal

Implementation of academic audit process

Academic Audit:

An **academic audit** is a systematic process designed to evaluate, assess, and improve the quality of academic practices, processes, and outcomes. The main objective is to ensure that academic standards are met, that the curriculum is relevant and effective, and that teaching, learning, and assessment practices align with the institution's goals and regulatory requirements. Academic audits help identify strengths, weaknesses, and areas for improvement in the educational system.

The academic audit will be conducted once a year by external academic experts. The list of academic experts from various institutions will be selected by the department, and the nominations will be made by the head of the institution.

- ❖ A **detailed audit schedule** will be prepared by the Academic Coordinator at least one week before the commencement of the audit. This schedule will be circulated to the department academic coordinator, head of the department, and faculty members.
- ❖ **An opening meeting** for the auditors will be conducted by the Academic Coordinator on the day the audit begins.
- ❖ The experts will review the academic processes using the **Academic Audit Checklist – THEORY (AC 26a)** and **Academic Audit Checklist – LABORATORY (AC 26b)**.



RAMCO INSTITUTE OF TECHNOLOGY

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NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Academic year 20... - 20... (Odd/Even Semester)

Academic Audit Check List -THEORY

Regulation - 2021

Date:

Name of the Academic Auditor (External):	
Designation	:
Institute/Industry address	:

Name of the Faculty member:

Department:

Course Code & Title:

Branch:

Course File : Course file/Canvas /Google classroom

1. Previous Academic Audit Comments and Action Taken Report: Yes ☐ No ☐
2. Availability of Academic Calendar: ☐ Adherence to Academic Calendar: ☐
3. Student Name list/ Syllabus Copy: ☐ Pre-Course Assessment (Question & Results): ☐
4. Notes of Lesson/Question Bank: Yes ☐ No ☐

Excellent		Good		Satisfactory		Need Improvement	
Suggestions for improvement:							

5. Class Time table ☐ Individual Time Table ☐ Log Book ☐
 Course Plan ☐ Course Assessment ☐ IAT Question Paper ☐
 IAT Mark Statements ☐ Theory Attendance ☐ Scheme of Evaluation and Answer Key ☐ (All documents as per ISO Format)
6. Tutorial Sheets (if applicable), Assignment Sheets / Individual Assignment Sheets /Case Study / Seminar/ Mini project, Rubrics for evaluation, IAT Answer Scripts samples:

Yes ☐ No ☐

Excellent		Good		Satisfactory		Need Improvement	
Suggestions for improvement:							

Figure 7.1: Academic Audit Checklist – Theory



RAMCO INSTITUTE OF TECHNOLOGY

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NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Academic year 20... - 20... (Odd/Even Semester) Academic Audit Check List - LABORATORY Regulation - 2021

Date:

Name of the Academic Auditor (External):	
Designation	:
Institute/Industry address	:

Name of the Faculty member:

Department:

Course Code & Title:

Course File : Course file/Canvas /Google classroom

1. Previous Academic Audit Comments and Action Taken Report: Yes ☐ No ☐
2. Availability of Academic Calendar: ☐ Adherence to Academic Calendar: ☐
3. Student Name list/ Syllabus Copy: ☐
4. Pre-Course Assessment (Question & Results): ☐
5. Lab Manual with Viva voce Questions: Yes ☐ No ☐

Excellent	☐	Good	☐	Satisfactory	☐	Need Improvement	☐
Suggestions for improvement:							

6. Class Time table ☐ Individual Time Table ☐ Log Book ☐
Course Plan ☐ Course Assessment ☐ Model Practical Exam Question Paper ☐
Mark Statement of Model Practical Exam ☐ Practical Marks Consolidated ☐
Practical Marks ☐ (All documents as per ISO Format)
7. Attendance & Presentation Marks: Yes ☐ No ☐
8. Internal assessments conducted for each course as per details provided in AU Clause 12.
Yes ☐ No ☐

Figure 7.2: Academic Audit Checklist – Laboratory

- ❖ Auditors will record their **suggestions for improvement** and overall performance in all checklists by verifying the course file.
- ❖ During the academic audit, all processes related to academics, based on the audit checklist, will be reviewed, including:
 - ✓ **Verification of the plan and execution of activities**
 - ✓ **Review of previous audit findings** and actions taken
 - ✓ **Identification of any risk factors** related to academics
- ❖ The audit process will conclude with a **closing meeting**, summarizing the audit outcomes.
- ❖ A **detailed academic audit report** will be prepared by the department academic coordinator, based on the good practices followed and the scope for improvement identified in the audited theory and laboratory course checklists.
- ❖ **Overall observations** and suggestions for improvement will be reviewed by the Academic coordinator, HOD and Principal.

Internal Audit:

An internal audit will be conducted twice, once after the Internal Assessment Test I and once during the vacation period. A detailed audit schedule, prepared by the management representative, will be circulated 15 days prior to the audit. The process will begin with an opening meeting, and any schedule changes will be communicated as needed. Each department's academic process will be audited for half a day to a full day by auditors from other departments. Auditors will record their observations in an audit notes document, covering areas such as plan execution, previous audit findings, quality objectives, risk identification, and performance comparison with the previous year and university exam results. The audit concludes with a closing meeting summarizing the findings. Auditors will submit a detailed report within a week, highlighting good practices, non-conformances, and improvement potential. A consolidated report will then be prepared by the management representative.

External Annual Surveillance Audit:

TUV Rhineland will send the audit schedule and plan 15 days before the audit date. An opening meeting will be held on the first day with management and process owners. Two auditors for one day, or one auditor for two days, will assess the ISO procedures and quality management system, focusing on the teaching and learning process and suggesting improvements. The closing meeting will summarize the audit findings, including good practices, observations or non-conformances, and improvement potential. A detailed report will be sent within a week, and corrective and preventive actions will be taken based on the audit outcome.

ACTION PLAN BASED ON AUDIT:

- ❖ Based on the Audit Report, the corrective actions and preventive measures are taken for Observations, Findings and Non Conformance (if any).
- ❖ In case of any Risk Identified during the Audit will be analysed and Preventive actions will be taken.
- ❖ The outcomes of Internal Audits and External Surveillance Audits will be reviewed in CAC Meeting, Department Review Meeting and Management Review Meeting.

Conducted Academic Audits:

In our department, 3 academic audits are conducted from 2021 to 2024. The audit observation, Action taken and Effectiveness are as follows:

Table 7.2 Academic Audit (2023-2024),(2022-2023),(2021-2022)

Observations	Action taken for Continuous Improvement	Effectiveness
2023-2024		
Good Practices:		
✓ Course Materials and Log books are well organized and maintained in all the aspects systematically. ✓ Mark split up in students record notebook are well organized and planned. ✓ Efforts taken for slow learners are appreciable. ✓ Innovative practices followed in TLP is appreciated. ✓ Virtual lab and NPTEL videos are planned well for laboratory courses.	Planned to continue the recorded good practices	• Continuous improvement in TLP. • Innovative practices in analytical courses helps to improve peer learning. • Providing student learning environment on cutting edge technologies.
Findings:		
Suggested to include hand written notes in unit wise study materials.	Prepared Hand written Notes for most of the problematic subjects.	Enhanced clarity and understanding for students.
Along with Assignment releasing date, submission date with faculty signature also can be added for evaluation and assessment purpose.	For better evaluation and assessment purpose, the assignment submission date has been included in the AC09b Assignment sheet.	Improved tracking and assessment of student submissions.
Models/demo videos can be included for better understanding and visualization of concepts.	Conducted demo session on the topic (Reciprocating Pump) in CE3401 – Applied Hydraulics Engineering subject.	Better understanding and visualization of concepts.
Field oriented Mini project and	Mini project in the topic	Increased student

can be included for better involvement and brainstorming among students.	Stabilization of soil were conducted in the laboratory course CE3413 – Soil Mechanics Laboratory.	involvement and brainstorming.
Individual CO and overall CO attainment to be updated for both semesters.	Individual CO and overall CO attainment were updated for both semesters.	Improved tracking of course progress.
Content beyond syllabus can be well planned and managed for some of the problematic courses.	Content beyond syllabus were identified and conducted in several domain subject.	Enriched student knowledge.
More students can be encouraged for doing more number of Online NPTEL courses.	More students will be encouraged for doing more number of Online NPTEL Courses in the upcoming semester.	Improvement in student pursuing online courses through Edx / Coursera / NPTEL platform.
Repeated lab exercises and doubt clearing sessions can be added to support weak students for better understanding.	Special class has been planned and conducted for the course CE3611 – Building Drawing and Detailing Laboratory for doubt clearing.	Improved support for weaker students.
Innovative practices planned in laboratory course plan to be conducted effectively.	For lab courses, innovative practices has been planned and conducted.	Innovative practices in laboratory courses helps to improve student engagement and learning.
Drone surveying and DGPS like advanced training can be taught for more exposure.	Professional Development Training on DGPS Technology has been conducted on 19.02.2024 to 23.02.2024 for IV Year Students, Faculty and Lab Technicians.	Increased exposure to advanced surveying techniques.
Suggested to include inference and interpretation of results with respect to standard values in lab manuals.	Inference and Interpretation of Results have been done in the all-laboratory courses.	Strengthened students' analytical skills.
2022-2023		
Good Practices:		
✓ All parts of the course materials' organisation and upkeep are excellent. ✓ The course plan has properly planned out. ✓ Manual, Log Book, and Student Records has been organized and planned systematically. ✓ TLP is acknowledged for its innovative techniques	Planned to continue the recorded good practices.	• Continuous improvement in TLP • Improvement in academic performance

✓ Giving students project individually is a good practice. ✓ Efforts taken for bright and slow learners are appreciable.		
Findings:		
Suggested to include mini projects/mini practical sessions for laboratory courses wherever required.	Practical sessions were conducted on the topics like Design and detailing of structures, Quality Assurance and Quality Control in Construction, Total Station & DGPS for students through Professional societies.	Improved practical knowledge and hands-on experience.
Suggested to give case studies/mini projects on sustainability.	Students have undertaken Household Waste Composter project and Real time case studies related to sustainability were discussed among students by the various experts from sustainability field through IGBC student chapter.	Promoted sustainability awareness.
Gate questions can be asked in Internal Assessment Tests.	Gate questions has been included in the Highway and Railway Engineering, Soil Mechanics, Water supply and Wastewater Engineering IAT questions.	Improved preparation for competitive exams.
Teaching Aids can be included in the course plan.	Instructional Methods and Teaching Aids were included in both course plan and log book in all theory and laboratory courses.	Enhanced teaching effectiveness through varied aids.
Suggested to include more tutorial and assignments for problematic papers.	Planned and conducted more tutorials and assignments for the problematic papers.	Continuous improvement in TLP.
Publication of students' project needs follow-up	5 students project has been published in SCI/Scopus/UGC journals.	Improved academic visibility and recognition.
Suggested to do the follow up measurable action for the activities implemented.	Academic quality policy has been revised as AC10b Rev.No.2 for Regulation 2021.	

2021-2022		
Good Practices:		
✓ Course file well maintained in all the aspects systematically. Course content are well organised. ✓ Document entry & signature completely filled. ✓ Lab manual file and log book well maintained.	Planned to continue the recorded good practices.	• Continuous improvement in TLP • Providing student learning environment on cutting edge technologies.
Findings:		
Beyond curriculum, topics can be taught if possible and to be documented.	Conducted Value added course on Open Building Designer and Skill development training programs on BIM Software were conducted beyond the Curriculum.	Broadened skill sets and industry exposure for students.
Field visit & real time inspection & Surveying can be initiated.	Industrial Visit on Ramco Industries Limited, Gangaikondan, L&T Construction Project – Cochin Dry Dock Site, Kerala and RDC Concrete (India) Pvt. Ltd, Cochin.	Provided real-world exposure to students.
	Field Visit and Real time Inspection were carried out by students on Mine surveying using Total Station in Pandalgudi and Preparation of topographical layout and bund profile for Kondeneri Kanmayee Lake's using DGPS & Drone, Rajapalayam.	Reinforced practical learning in real-world contexts.
Beyond curriculum, case studies and site visit can be initiated for better understanding.	Beyond curriculum 34 students have completed the L&T EduTech online courses like Concreting Techniques and Practices, Design of Reinforced Concrete Buildings & Practice, and Formwork Engineering Practices.	Enhanced student understanding of industry practices.
During laboratory courses, students may be demonstrated with standard operating procedure.	For the benefit of students, codal references for all experiment has been prepared for every	Standardized practices and reinforced technical knowledge.

	laboratory course and is displayed in each lab.	
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Conducted Internal Audits:

In our department, 7 internal audits are conducted from 2021 to 2024. The date of auditing, auditors and report are as follows.

Internal Auditing- 1 (2021-2022)

Table 7.3 Internal Auditing - 1 (2021-2022)

Date of Auditing	Auditors	Report
ISO Audit 30.10.2021	Dr.M.Gomathy Nayagam, ASP/CSE Mrs.N.Nithya, AP/CSE Mr.N.Ganesh, AP(SG)/EEE	1.PO-PSO mapping can be done for the events conducted by the professional society 2.Opinion of the HOD has to be filled in the AC02a –Faculty subject preference
Action Taken: 1. PO-PSO mapping was included in all the event circular conducted through Professional societies and Department Association. 2. The opinion of the HOD in AC02a –Faculty subject preference was updated.		

Internal Auditing- 2 (2021-2022)

Table 7.4 Internal Auditing - 2 (2021-2022)

Date of Auditing	Auditors	Report
ISO Audit 26.05.2022	Dr.A.Azhagu Jaisudhan Pazhani, ASP/ECE Mr.R.Arun Kumar, AP/Mech, Dr.U.Kumaran, AP(SG)/Maths	1.More students can be encouraged do online courses 2.Minutes can be drafted in such a way that, retrieval of information is ease
Action Taken: 1.Most of the students were completed the online course in various platforms like L&T Edutech, Imarticus, Coursera etc., 2.All meeting minutes were written so that information could be easily retrieved.		

Internal Auditing- 3 (2022-2023)

Table 7.5 Internal Auditing - 3 (2022-2023)

Date of Auditing	Auditors	Report
ISO Audit 02.11.2022	Dr.A.Azhagu Jaisudhan Pazhani, ASP/ECE Mr.R.Arun Kumar, AP/Mech, Dr.U.Kumaran, AP(SG)/Maths	1. Only IAT based performance is alone used for categorizing students. It can be penetrated in further aspects including other direct assessments, assignment/tutorial performance, classroom assessments and attendance.

		2. Faculty members can be motivated to pursue online course based on the domain their having expertise which can be helpful for them to get domain certification in any platform including NPTEL and Coursera. 3. The same can be suggested to students also. 4. Students can be do projects on other domains 5. Insisted to prepare consolidated document for placement and training, Research activities, publication in each year
Action Taken: 1. Faculty members were completed the online courses based on their domain and students also done online courses in various platform like NPTEL, Coursera etc., 2. Few students had done projects in various domain such as Structural, Environmental, Infrastructure, Geotechnical etc., 3. Consolidated document for placement and training and Research activities for each year were updated.		

Internal Auditing- 4 (2022-2023)

Table 7.6 Internal Auditing - 4 (2022-2023)

Date of Auditing	Auditors	Report
ISO Audit 16.02.2023	Mr.R.Arun Kumar, AP/Mech, Mrs.M.Swarna SudhaAP(SG)/CSE	1. Feedback for the VAC course has to be collected from the students along with report of each course. 2. Internship report has to be collected from the students and the review has to be conducted as part of assessment. 3. Appointment of Adjunct Faculty is initiated, yet to be taken further process. Website has to be updated at regular interval
Action Taken: 1. VAC Feedback were collected from the students along with report. 2. Collected the internship report and the review were conducted as part of assessment. 3. Appointment of Adjunct Faculty was completed. 4. Updated the department activities in website.		

Internal Auditing- 5 (2022-2023)**Table 7.7 Internal Auditing - 5 (2022-2023)**

Date of Auditing	Auditors	Report
ISO Audit 15.06.2023	Dr.L.Sathikala ASP/Maths Mrs.G.Gnanapriya. AP(SG)/ECE	1. In CE3403 Concrete technology course, pre assessment marks, assignment question & marks, AC10b form needs to be updated 2. For R2021, in course plan the weightage given for End Semester and Internal Assessment is not as per the form AC06a for few courses. 3. Internship Details has to be collected from the students and consolidation is missing.
Action Taken: 1. Pre assessment marks, assignment question & marks, AC10b form were updated in CE3403 Concrete technology course. 2. In course assessment form AC06a, all the courses followed the same weightage for End Semester and Internal Assessment Test. 3. Student Internship Details were collected and consolidated the same.		

Internal Auditing- 6 (2023-2024)**Table 7.8 Internal Auditing - 6 (2023-2024)**

Date of Auditing	Auditors	Report
ISO Audit 27.11.2023	Dr.M.Swarna Sudha AP(SG)/CSE Dr.J.Jerold John Brito AP(SG)/Mech Mr.B.Kannan, AP(SG)/ECE	Suggested to organise subject-specific events for CO-PO mapping calculations.
Action Taken: Conducted subject-specific events for CO-PO Mapping calculations.		

Internal Auditing- 7 (2023-2024)**Table 7.9 Internal Auditing - 7 (2023-2024)**

Date of Auditing	Auditors	Report
ISO Audit 19.06.2024	Mrs.V.Srirenga Nachiyar. AP(SG)/ECE Mr.K.Vignesh Saravanan AP(SG)/CSE	1. CE3601- Design of Steel Structural Elements – Proof can be included for the lecture delivered by Alumni. 2. CE3602 Structural Analysis II Content beyond syllabus from DAC members proof has to be included. 3. Placement – Need to prepare consolidated with current mobile number of the student

		and collect the appointment order. 4. Suggested to tie up with the industry to conduct VAC and improve placement through internship. 5. Suggested to do more student online courses.
Action Taken: 1. Event report and proof documents were updated. 2. Content beyond syllabus from DAC members proof was included. 3. Prepared the consolidated data along with the appointment order and current mobile number were updated. 4. Conducted VAC along with industry and improved the placement through industry. 5. More number of II & III year completed the online course through NPTEL, Coursera etc.,		

External Audit Comments for college:

Three External audits were conducted from 2021 to 2024 for continuous improvement are given to college. The date of auditing, auditors and report are as follows.

Academic Year: 2021-2022

Table 7.10 External Audit (2021-2022)

Date of Auditing	Auditors	Audit points
19.01.2022 & 20.01.2022	Dr.Viswanathan V TUV Rheinland(India) Pvt.Limited,Coimbatore	Positive Findings: Support and continual improvement focus by the management. Improvement Potential: Improvement in capturing student data from ERP to be done.
Actions taken: ERP processes are updated to collect students' data like elective preference and final elective list, feedback collection etc., from students.		

Academic Year: 2022-2023

Table 7.11 External Audit (2022-2023)

Date of Auditing	Auditors	Audit points
08.12.2022 & 09.12.2022	Dr.Viswanathan V TUV Rheinland(India) Pvt.Limited,Coimbatore	Positive Findings: 1. The top management and leadership team's strategic direction for strengthening the system and process. 2. Implementation of every internal audit observation from the past. 3. Participation of the auditee team in the audit

		and transparency. 4. This year's successful admissions and newly started courses are applauded. Improvement Potential: 1. Micro level planning, Execution of contents may also be reviewed strongly. Sample: Theory – Problems. 2. During the execution of practical subjects' evaluation on the core objective may be done. 3. Allocation of hours for a subject during individual timetable may always be without decimal value. 4. Class committee meeting, Disciplinary process in the department may adhere to its respective regulations.
Actions taken: 1.New Course Plan design work initiated. 2.Sharing hours in Timetable is removed 3.CCM/ CoCM meetings conducted and recorded as per ISO standards.		

Academic Year: 2023-2024

Table 7.12 External Audit (2023-2024)

Date of Auditing	Auditors	Audit points
26.12.2023 & 27.12.2023	1.Dr.Viswanathan V TUV Rheinland(India) Pvt.Limited,Coimbatore	Positive Findings: 1. Achievements by the college including academics, sports, research and overall rating has been excellent this year 2. Budget allocated to all departments including library and its usage 3. Support and transparency by the auditee team was excellent. Improvement Potential: 1. Documents and execution transparency may be given to parent department on subjects handled by other departments 2. Scrutiny of the internal examination question papers to be taken care by the respective department.
Actions taken: 1. The parent department is provided with complete access to relevant documents and execution details for subjects handled by other departments to enhance oversight and coordination. 2. The responsibility for the thorough review and approval of all internal examination question papers before submission to the exam cell has been assigned to the respective department HOD.		

7.3. Improvement in Placement, Higher Studies and Entrepreneurship (10)

Placement, higher studies and Entrepreneurship have distinct focus of students. The department frequently organizes various events for the development of students in all aspects. The detailed statistics of placement, higher studies and entrepreneurs are given in the table 7.13.

Table.7.13 Details of Placement, Higher studies and Entrepreneurship

Item	CAY m1 (2023-2024)	CAYm2 (2022-2023)	CAYm3 (2021-2022)
Total No. of Final Year Students (N)	21	17	22
No. of students placed in companies or Government Sector (x)	15	12	11
No. of students admitted to higher studies with valid qualifying scores (GATE or equivalent State or National Level Tests, GRE, GMAT etc.,) (y)	1	1	3
No. of students turned entrepreneur in engineering/technology (z)	1	-	-
$x + y + z =$	17	13	14

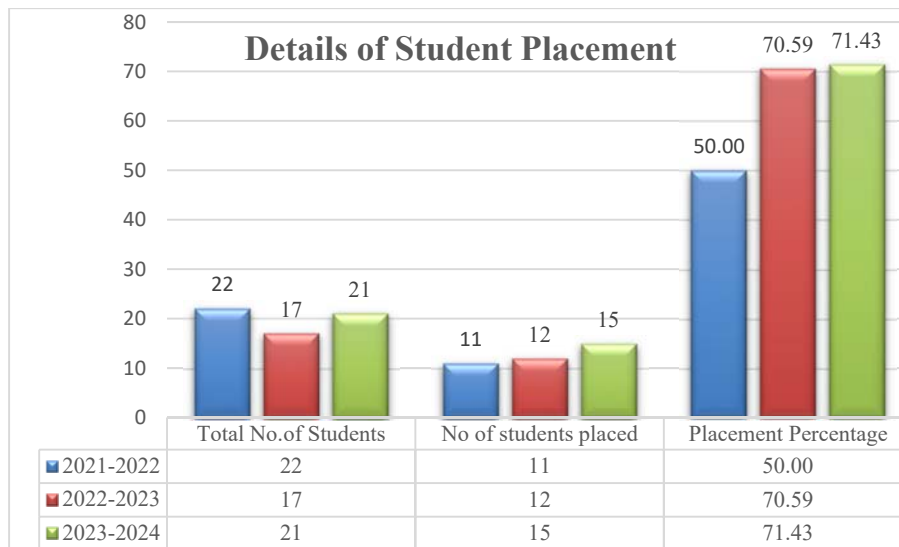


Figure 7.3 Details of Student Placement

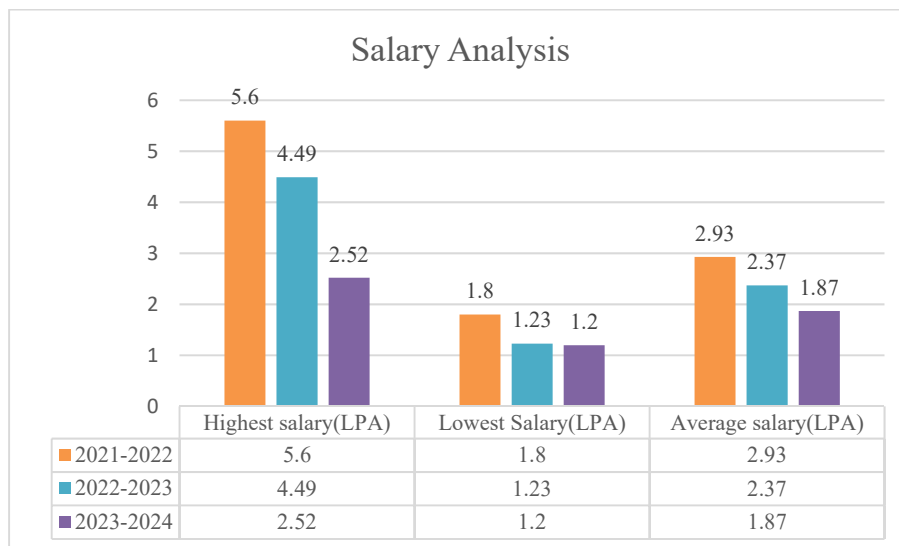


Figure 7.4 Placement Comparison with Salary Analysis

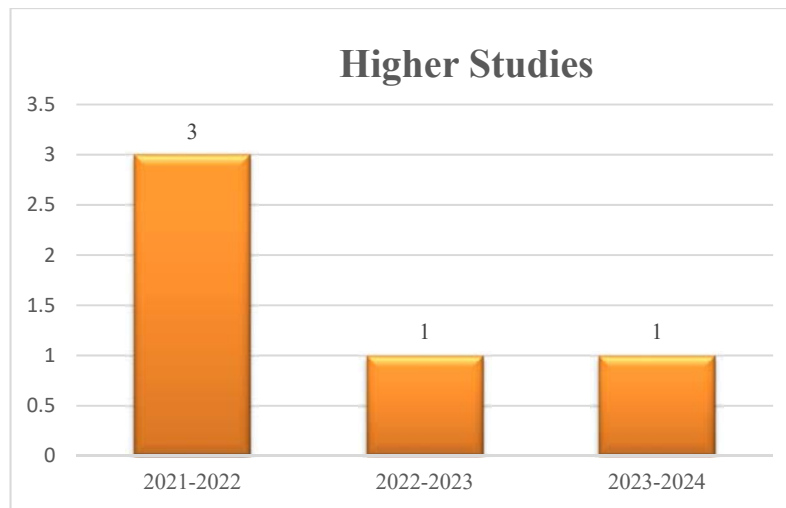


Figure 7.5 Details of Higher Studies

Table 7.14 Salary break-up details

Category	CAYm1 (2023-2024)	CAYm2 (2022-2023)	CAYm3 (2021-2022)
Total No. of Final Year Students	21	17	22
No. of Students Placed	15	12	11
No. of students in the salary range 1 – 2 LPA	9	6	4
No. of students in the salary range 2 – 3 LPA	6	3	3
No. of students in the salary range 3 – 4 LPA	Nil	2	1
No. of students in the salary range 4 – 5 LPA	Nil	1	1
No. of students in the salary range 5 – 6 LPA	Nil	Nil	2

Table 7.15 (2020 – 2024) Batch Placement details

Sl. No.	Name of the Company	Core / Software	Number of Students Placed	Salary (LPA)
1	Ambal Construction	Core	1	1.44
2	Ayothi Consultancy	Core	2	1.98
3	Pramoda Rebar Private Limited	Core	1	1.20
4	Pinnacle Infotech Solutions	Core	3	2.70
5	Absara Consultancy	Core	2	2.04
6	RCI Digital Solutions	Core	1	2.52
7	Sivam Builders	Core	2	1.80
8	RAROVAH Engineering PVT.LTD	Core	1	1.92

9	Shri Vetri Builders	Core	1	1.68
10	Balaji Construction	Core	1	1.44
Highest Salary (in LPA)				2.52
Lowest Salary (in LPA)				1.20
Average Salary (in LPA)				1.87

Table 7.16 (2019 – 2023) Batch Placement details

Sl. No.	Name of the Company	Core / Software	Number of Students Placed	Salary (LPA)
1	R.K.M Powergen Pvt.Ltd	Core	1	3.6
2	Pinnacle Infotech Solutions	Core	2	2.40
3	The Ramco Cements Limited	Core	1	3.50
4	Jasper International Engineering Consultants	Core	1	1.23
5	The Ramco Cements Limited	Core	1	4.49
6	SANSONS-Civil Construction	Core	1	1.80
7	Ambal Construction	Core	1	1.44
8	P.P Builders	Core	1	1.80
9	Shri Vetri Builders	Core	1	1.68
10	RD ASSOCIATES	Core	1	2.16
11	River Construction & Consultancy	Core	1	1.98
Highest Salary (in LPA)				4.49
Lowest Salary (in LPA)				1.23
Average Salary (in LPA)				2.37

Table 7.17 (2018 – 2022) Batch Placement details

Sl. No.	Name of the Company	Core / Software	Number of Students Placed	Salary (LPA)
1	Absara Consultancy	Core	1	2.04
2	The Ramco Cements, Limited	Core	1	5.03
3	The Ramco Cements, Limited	Core	1	5.60
4	Maruthi Builders	Core	1	1.80
5	AEON Integrated Building Design LLP (“AIBDC”)	Core	1	4.66
6	RiDh Engineering Services	Core	1	3.41

7	Innowell Engineering International Pvt Ltd.	Core	1	2.22
8	Sivam Builders	Core	1	1.80
9	SANSONS-Civil Construction	Core	1	1.80
10	RD ASSOCIATES	Core	1	2.04
11	River Construction & Consultancy	Core	1	1.80
Highest Salary (in LPA)				5.60
Lowest Salary (in LPA)				1.80
Average Salary (in LPA)				2.93

Table 7.18 Higher Studies details

Year	Number of students enrolling into higher education	Student Name	Program to be graduated from	Name of Institution Joined
CAYm1 (2023-2024)	1	Dheena Dhayalan R	M.E. Structural Engineering	Kumaraguru College of Technology, Coimbatore
CAYm2 (2022-2023)	1	Hari Hara Pandiyan V	M.E. Structural Engineering	Mepco Schlenk Engineering College, Sivakasi
CAYm3 (2021-2022)	3	Jenitha G	M.E. Structural Engineering	Thiagarajar College of Engineering, Madurai
		Meenaa Bharathi S	M.E. Construction Management	Thiagarajar College of Engineering, Madurai
		Sudhakaran A	M.E. Structural Engineering	Alagappa Chettiar Government College of Engineering and Technology, Karaikudi,

Table 7.19 Entrepreneur details

Sl.No.	Name of the student Entrepreneur	Year of passed out	Type of Business	Designation
1	S.Arun Kumar	2024	Family Business	Partner

7.4. Improvement in the quality of students admitted to the program**(10)**

Assessment is based on improvement in terms of ranks/score in qualifying state level/national level entrances tests, percentage marks in Physics, Chemistry and Mathematics in 12th Standard and percentage marks of the lateral entry students.

Item		CAY (2024 – 25)	CAYm1 (2023 – 24)	CAYm2 (2022-23)	CAYm3 (2021 – 22)
National Level Entrance Examination (Name of the Entrance Examination)	No. of Students admitted	0	0	0	0
	Opening Score/Rank	0	0	0	0
	Closing Score/Rank	0	0	0	0
State/University/Level Entrance Examination/Others (Tamil Nadu Engineering Admission)	No. of Students admitted	38	43	31	42
	Opening Score/Rank	179.5	172	178	189.28
	Closing Score/Rank	91.5	92	80.5	137.46
Name of the Entrance Examination for Lateral Entry or lateral entry details	No. of Students admitted	3	4	3	11
	Opening Score/Rank	96	90.14	74.79	86
	Closing Score/Rank	77	68.29	66.62	65.71
Average CBSE/Any other Board Result of admitted students (Physics, Chemistry & Maths)		63	65	64	82